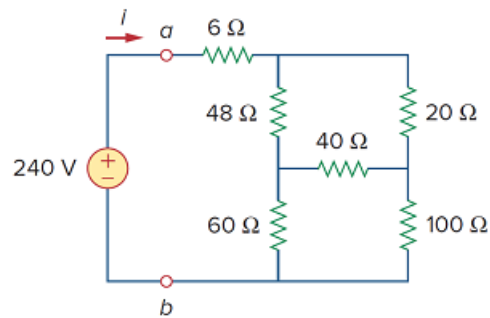


Problem

For the bridge network in Fig. 2.54, find R_{ab} and i .

Figure 2.54:



Step-by-step solution

Step 1 of 9

Refer to Figure 2.54 in the textbook.

To find the resistance seen by the voltage source, R_{ab} , first convert the Y network formed by the resistances $48\ \Omega$, $60\ \Omega$ and $40\ \Omega$ into Δ network. Choose the resistances as follows:

$$R_1 = 60\ \Omega$$

$$R_2 = 48\ \Omega$$

$$R_3 = 40\ \Omega$$

Calculate the equivalent resistances in the Δ network.

$$\begin{aligned} R_a &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \\ &= \frac{(60)(48) + (48)(40) + (40)(60)}{60} \\ &= 120\ \Omega \end{aligned}$$

$$\begin{aligned} R_b &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \\ &= \frac{(60)(48) + (48)(40) + (40)(60)}{48} \\ &= 150\ \Omega \end{aligned}$$

$$\begin{aligned} R_c &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \\ &= \frac{(60)(48) + (48)(40) + (40)(60)}{40} \\ &= 180\ \Omega \end{aligned}$$

Step 2 of 9

Draw the modified circuit diagram.

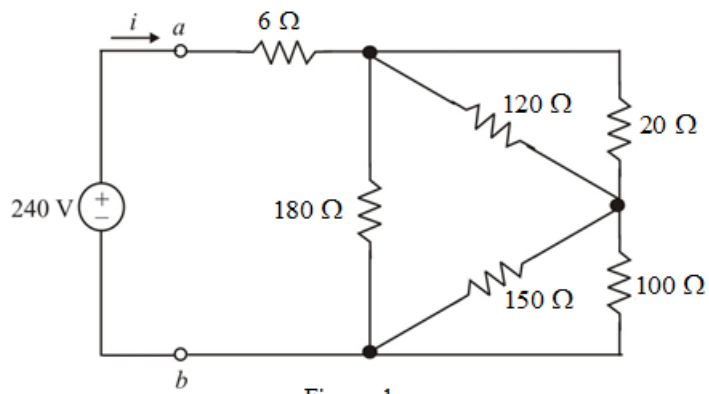


Figure 1

Step 3 of 9

In Figure 1, the resistances $120\ \Omega$ and $20\ \Omega$ are connected parallel. Find its equivalent resistance.

$$120\ \Omega \parallel 20\ \Omega = \frac{(120)(20)}{120 + 20} \\ = 17.143\ \Omega$$

Similarly, the resistances $150\ \Omega$ and $100\ \Omega$ are connected parallel. Find its equivalent resistance.

$$150\ \Omega \parallel 100\ \Omega = \frac{(150)(100)}{150 + 100} \\ = 60\ \Omega$$

Step 4 of 9

Draw the modified circuit diagram.

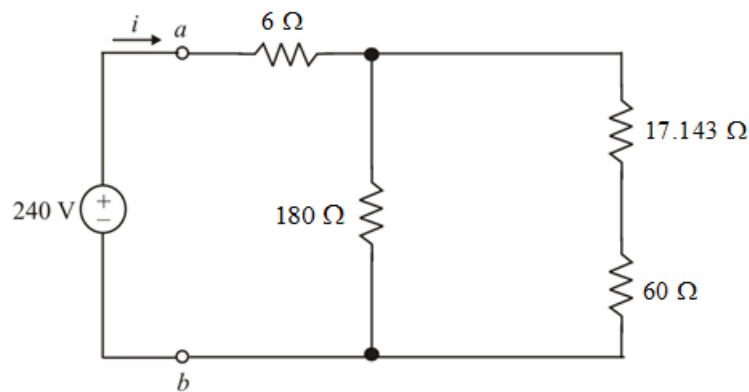


Figure 2

Step 5 of 9

In Figure 2, the resistances $17.143\ \Omega$ and $60\ \Omega$ are connected in series. Find its equivalent resistance.

$$17.143\ \Omega + 60\ \Omega = 77.143\ \Omega$$

This $77.143\ \Omega$ resistor and $80\ \Omega$ resistors are in parallel. Their combined resistance is:

$$77.143\ \Omega \parallel 180\ \Omega = \frac{(77.143)(180)}{77.143 + 180} \\ = 54\ \Omega$$

Step 6 of 9

Draw the circuit modified diagram to find the equivalent resistance seen by the source.

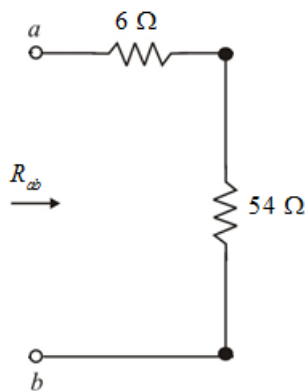


Figure 3

Step 7 of 9

Find the resistance R_{ab} using Figure 3.

$$R_{ab} = 6\ \Omega + 54\ \Omega \\ = 60\ \Omega$$

Therefore, the resistance seen by the source, R_{ab} is $\boxed{60\ \Omega}$.

Step 8 of 9

Place the equivalent resistance R_{ab} across the terminal of a and b .

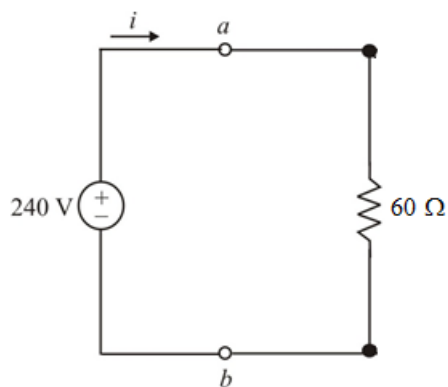


Figure 4

Step 9 of 9

Calculate the current i using Ohm's law.

$$\begin{aligned} i &= \frac{v}{R} \\ &= \frac{240}{60} \\ &= 4 \text{ A} \end{aligned}$$

Therefore, the value of current i is 4 A.